A Finite-Rate Selection Law for Annealed Gibbs Sampling

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Abstract

A sampling algorithm can generate plausible examples from a probability model but in the wrong proportions. We study this error in annealed Gibbs sampling, which updates individual variables while lowering the temperature. In the Curie–Weiss (CW) model, binary variables tend to share one sign. A weak field favors $+1$ but competes with random fluctuations as cooling chooses the sign. Classical theory predicts this choice; comparing it with the equilibrium probabilities gives a law for how much of the field's preference survives. Much slower cooling raises the surviving preference only modestly. The law agrees closely with numerical evolution of the exact CW chain. In a Hopfield model with several stored patterns, we remove runs in one mode after brief cooling and reheat the rest so that they choose again. Competition amplifies the preference between a pattern and its negative. A Gaussian approximation captures this effect, with discrepancies across pattern counts, sizes, and update rules. The results relate the sign chosen during cooling to final mode probabilities; the CW calculation is a reference for comparing cooling schedules.

1. Introduction

Suppose a model of handwritten digits assigns twice as much probability to "3" as to "8". A sampling algorithm should then produce about twice as many "3"s as "8"s over many runs. It can produce recognizable examples of both digits and still get this proportion wrong. Probabilities and averages estimated from these samples can then be biased. How does the sampling algorithm change these proportions?

We study annealed Gibbs sampling. A Gibbs update redraws one variable from its conditional distribution given the other variables. High temperature gives a broad distribution; lower temperature concentrates probability around likely outcomes. Cooling, or annealing, lowers the temperature to its target value. During cooling, moving between groups of likely outcomes becomes difficult. Further updates within one group need not correct how often each group was chosen [9].

To isolate this error from inaccuracies in a learned model, we use the Curie–Weiss (CW) model. Its binary variables tend to align, so cooling makes most of them share one sign. A weak field favors the positive sign but competes with random fluctuations while cooling selects the sign. Both the model's probabilities and the evolution of the sampling algorithm can be computed from the number of positive variables alone.

We connect a classical prediction of this sign choice [6, 7, 10] to the error in final proportions. The resulting law quantifies how much of the field's preference survives cooling and shows why

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much slower cooling improves it only modestly. Extending the argument to a Hopfield model with several stored binary patterns predicts that competition among patterns strengthens this preference. We test the prediction on runs reheated to choose a mode again.[1] It closely matches three-pattern simulations but misses by up to 11% of the prediction at other pattern counts or sizes.

2. An exact model of mode selection

A configuration is a vector $x = (x_1, \ldots, x_N)$ of binary variables $x_i \in \{-1, +1\}$, called spins. The CW model assigns higher probability to configurations in which spins agree:

$$\pi_{\beta,G}(x) \propto \exp\left[\frac{\beta}{2(N-1)} \sum_{i \neq j} x_i x_j + \beta G \sum_i x_i\right]. \tag{1}$$

At $\beta = 0$, all configurations are equally probable. Increasing β strengthens the preference for alignment and corresponds to cooling; β is therefore the *inverse temperature*. The *field* G favors spins of its sign and stays fixed during each cooling run. At fixed temperature, Gibbs updates eventually reproduce $\pi_{\beta,G}$, the *equilibrium distribution*.

The average spin, $m = N^{-1} \sum_i x_i$, is the *magnetization*. Its sign separates mostly positive from mostly negative configurations. Without a field, in the limit $N \to \infty$, the equilibrium magnetization concentrates at zero for $\beta < 1$ and at two opposite nonzero values for $\beta > 1$, with the *transition* at $\beta = 1$. The configurations around these two values form two *modes*, or groups of likely outcomes.

For CW, each Gibbs update chooses a spin uniformly. Updates occur at random times at rate N; one unit of time, N updates on average, is an *expected sweep*. All configurations with the same number k of positive spins have the same rates of changing k. We can therefore track $N + 1$ probabilities, one for each $k = 0, \ldots, N$, instead of 2^N configurations [8], and evolve them numerically (Appendix A).

Let q_{chain} be the probability that the algorithm ends with $m > 0$, and q_{eq} the probability of $m > 0$ under the final equilibrium distribution, both conditioned on $m \neq 0$. For a weak field, sampled and equilibrium log-odds are both approximately proportional to G (Section 3). Their ratio measures the surviving preference:

$$c = \frac{\operatorname{logit} q_{\text{chain}}}{\operatorname{logit} q_{\text{eq}}}, \qquad \operatorname{logit} q = \log \frac{q}{1-q}, \tag{2}$$

for $q_{\text{eq}} \neq 1/2$. $c = 1$ reproduces the equilibrium preference, whereas $c = 0$ gives equal mode probabilities. If equilibrium odds are 4:1, then $c = 0.2$ gives sampled odds of $4^{0.2}$:1 $\approx$ 1.32:1.

3. How much of the preference survives?

The sign chosen near $\beta = 1$ can determine the final mode probabilities. There, a small random imbalance between positive and negative spins can grow into either mode (Figure 1a). The field biases this early choice. After the modes form, switching between them can take much longer than settling within either one [5, 8].

1. Code, saved runs, the full protocol, and material omitted here (relaxation times, the full $p = 2$ calculation, an MNIST illustration): https://github.com/ali-izhar/finite-rate-selection-law.

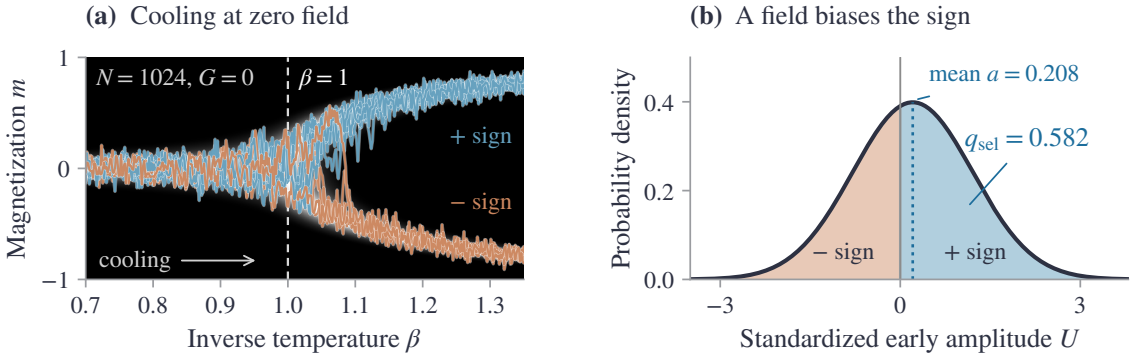


Figure 1: **How cooling selects a sign.** (a) Fourteen zero-field paths ($N = 1024$) of an approximate drift-plus-noise equation with the full drift $-m + \tanh(\beta m)$, not its approximation $(\beta - 1)m$ (Appendix A; Table 5) split into two modes, colored by final sign. Brightness shows the exact equilibrium probability relative to its peak at each temperature. (b) Analytical density, not a fit to (a) ($N = 2048$, $\gamma = 0.001$, $G = 0.0006125$, the largest Hopfield field of Section 4; $a = 0.208$): $U \sim \mathcal{N}(a, 1)$. The blue area is $q_{\text{sel}} = \Phi(a)$.

Near $m = 0$ and $\beta = 1$, the mean rate of change of m, or *drift*, is approximately $(\beta - 1)m + G$. The first term reduces an imbalance below $\beta = 1$ and amplifies it above. Random Gibbs updates add *noise*, with variance approximately $2/N$ per expected sweep (Appendix A). Let $\beta(t) = 1 + \gamma t$ increase at a constant rate $\gamma > 0$ and cross the transition at $t = 0$. The growth exponent of m, $\int_0^t \gamma u \, du = \gamma t^2/2$, reaches order one after a time T of order $\gamma^{-1/2}$. In this time, the field contributes an imbalance of order GT, while the noise accumulates as $\sqrt{T/N}$ [3]. Their ratio is of order $G\sqrt{N}\,\gamma^{-1/4}$. Slower cooling therefore favors the field.

Dividing m by the positive growth factor $e^{\gamma t^2/2}$ of the linear drift preserves its sign and gives the early *amplitude*: the accumulated field plus noise. In the linear approximation, the noise is a weighted sum of independent Gaussian increments. The amplitude is therefore Gaussian; dividing it by its standard deviation gives U, whose mean a measures the field relative to the noise. A classical calculation [6, 7, Appendix A] gives the probability q_{sel} of selecting $m > 0$:

$$q_{\text{sel}} \simeq \Phi(a), \qquad a = G\sqrt{N}(\pi/\gamma)^{1/4}, \tag{3}$$

where Φ is the standard Gaussian distribution function (Figure 1b). Without later switching, q_{sel} also gives q_{chain}.

When $|a| \ll 1$, both signs are nearly equally likely, $\operatorname{logit} \Phi(a) \approx 4a/\sqrt{2\pi}$, and sampled log-odds grow as $G\sqrt{N}\,\gamma^{-1/4}$. In equilibrium, reversing all N spins changes the field term of the log-probability by order NG. At a fixed final temperature, equilibrium log-odds thus grow as NG. Dividing gives

$$c \propto (N^2\gamma)^{-1/4}. \tag{4}$$

Sixteenfold longer cooling doubles c in this regime, and fourfold larger N halves it.

The *selection law*, Eq. (3) converted to c by Eq. (2), assumes a slow ramp ($\gamma \ll 1$), a start well below $\beta = 1$, and little switching after selection. A sufficient condition for the noise to stay below

$|m| \sim \gamma^{1/4}$, where growth stops being linear, is $N\gamma \gg 1$ [3] (Appendix A). At fixed N, the law does not extend to arbitrarily slow cooling: long sampling restores the equilibrium mode probabilities, but at $N = 2048$ a zero-field switch between modes takes on average about 5×10^7 expected sweeps at $\beta = 1.1$ (Appendix A).

Exact check. The selection law agrees with numerical CW evolution over the whole cooling run. Runs start from equilibrium at $\beta = 0.7$ and cool to 1.6 with $G = 0.0006125 \times 2048/N$; fixed NG keeps the equilibrium log-odds nearly constant across sizes. For $N = 1024$–4096 and $\gamma = 0.00025$–0.004, the numerical c exceeds the prediction in all nine cases, by 0.23–1.81% (Figure 2a; Table 2). Agreement holds even at the smallest tested $N\gamma = 0.256$, where $N\gamma \gg 1$ fails.

4. Competition among stored patterns

Hopfield model. A Hopfield model stores p patterns $\xi^\mu \in \{-1, +1\}^N$ with independent equiprobable entries. A configuration x has probability proportional to $e^{-\beta E_G(x)}$, with *energy*

$$E_G(x) = -\frac{1}{2N} \sum_{i \neq j} \sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu x_i x_j - G \sum_i \xi_i^1 x_i. \tag{5}$$

At low temperature and $p \ll N$, configurations concentrate near a stored pattern or its negative [1]. The *overlap* $m^\mu = N^{-1} \sum_i \xi_i^\mu x_i$ measures this resemblance: it is $+1$ at pattern μ and -1 at its negative. Each sample is labeled $(\mu, \pm)$ by the pattern with largest $|m^\mu|$ and the sign of that overlap. Here q_{chain} and q_{eq} are positive-sign probabilities among samples assigned to pattern 1, not pattern frequencies.

Competition with other patterns strengthens the field's preference for pattern 1 over its negative. Suppose the early overlaps are independent Gaussians with a common variance, the field shifts only the mean of m^1 (as in the CW model), and the largest $|m^\mu|$ determines the final pattern. For a shifted Gaussian, the sign log-odds at a given magnitude grow linearly in that magnitude. Winning selects larger magnitudes and so strengthens the sign preference. To first order in the field, averaging over winning magnitudes relative to one pattern (the CW case) gives

$$A_p = \frac{\mathbb{E} \max_{1 \leq \mu \leq p} |Z_\mu|}{\mathbb{E}|Z|}, \qquad c_{\text{pred}} = A_p c_{\text{CW}}, \tag{6}$$

where Z and the Z_μ are independent standard Gaussians and c_{CW} is c from exact CW evolution of the Hopfield cooling protocol (next paragraph), not of a linear ramp. The factors are $A_1 = 1$, $A_2 = \sqrt{2}$, $A_3 = 1.662$, and $A_4 = 1.836$ (Appendix C). The prediction also assumes equal Hopfield and CW equilibrium sign preferences; a small p/N guarantees neither this equality nor independent overlaps.

Reheating after a filtered start. Reheating lets a run that settled away from the favored mode choose again (*reselection*). We test the prediction here, not on fresh cooling, so Eq. (4) does not apply directly. A *permutation sweep* updates every spin once in a fresh random order (N updates, like an expected sweep). The *cooling schedule* has ten inverse temperatures from 0.6 to 1.6; each dwell is six relaxation times of the zero-field CW chain, across both signs for $\beta \leq 1$ and within one sign above (Table 3); in one relaxation time, the slowest deviation from equilibrium shrinks e-fold. Its 834 sweeps, computed at $N = 2048$, are reused at every size.

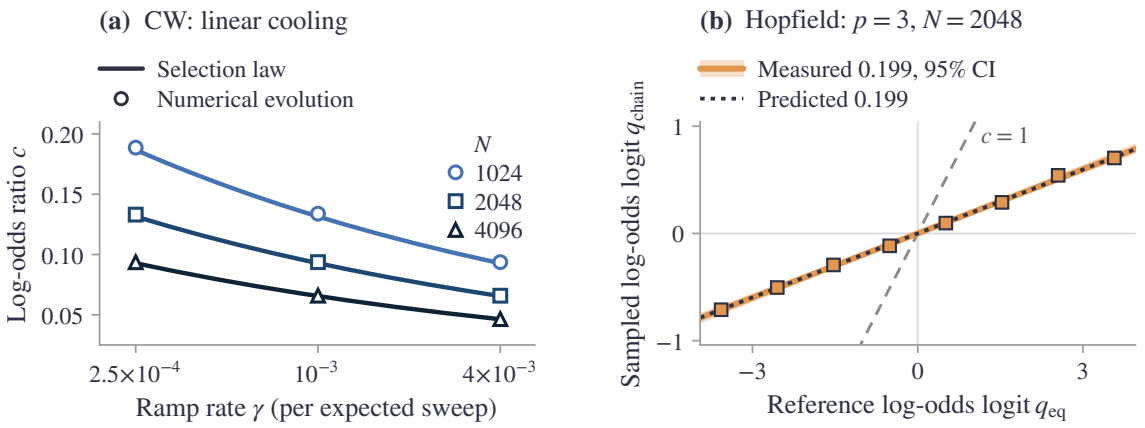


Figure 2: **How much of the equilibrium preference survives?** (a) CW ramps at fixed NG: $16\times$ slower cooling doubles c; the law is within 2%. (b) Hopfield samples retain about a fifth of the equilibrium log-odds. Each point pools twenty pattern sets; error bars are smaller than the symbols. The orange 95% band of the fitted slope c excludes reference-model uncertainty and contains the dotted prediction $A_3 c_{\mathrm{CW}}$; the dashed line $c = 1$ marks equilibrium.

Each run has three stages. First, a brief cooling through the schedule at two sweeps per temperature; removing runs labeled $(1, +)$ leaves a *filtered start*; for $G > 0$, it imitates a sampler that settled away from the favored mode. Second, a *reheat* for 30 sweeps at $\beta = 0.7$, where the drift reduces an imbalance. Third, the *final cooling* through the schedule. The removal also applies at $G < 0$, so fields $\pm G$ follow one protocol. From a sign-symmetric start, $q_{\mathrm{chain}}(G) + q_{\mathrm{chain}}(-G) = 1$; memory of the removal could break this. The measured sums are consistent with 1 but do not establish equivalence to fresh starts (Table 4). CW evolution of this protocol gives $c_{\mathrm{CW}} = 0.120$ at $N = 2048$; scaling final cooling by 0.25 to 4 (sixteenfold) raises it 1.48-fold, not the twofold of Eq. (4) (Appendix A).

Result. For three patterns and $N = 2048$, the positive sign occurs in 66.9% of samples assigned to pattern 1 at $G = 0.0006125$, against 97.26% in the equilibrium reference; their logits give $c = 0.70/3.57 \approx 0.20$. Unlike CW, this reference is approximate: its log-odds $2\beta_f NG\bar{m}^1$ use the zero-field mean positive-mode overlap $\bar{m}^1$ at the final β_f (Appendix B).

At all eight nonzero fields of the *field grid*, the sampled log-odds are approximately proportional to the reference log-odds (Figure 2b). Slopes fitted to twenty independently drawn pattern sets, weighted by the number of samples assigned to pattern 1, average to $c = 0.199$ (95% interval $[0.1908, 0.2095]$ from resampling pattern sets). Equal pattern-set weights give 0.2101, just above this interval; three other estimators give 0.1994–0.1997 (Appendix B). The prediction $A_3 c_{\mathrm{CW}} = 0.199$ uses no Hopfield samples and agrees closely, but approximation errors may cancel (Section 5).

5. Where the prediction departs

Other settings miss the prediction by up to 11% (Table 1; for $p = 2$, see Appendix C). They lack a Hopfield reference, so they use the exact CW reference and, assuming $q_{\mathrm{chain}}(G) + q_{\mathrm{chain}}(-G) = 1$,

Table 1: Predictions $A_p c_{\mathrm{CW}}$ use $c_{\mathrm{CW}} = 0.171, 0.120, 0.090$ at $N = 1024, 2048, 4096$; 95% intervals resample pattern sets. All rows use $|G| = 0.0006125$ (not rescaled with N).

p	N	predicted c	measured c
1	2048	0.120	$0.127\ [0.124, 0.131]$
2	2048	0.169	$0.188\ [0.184, 0.191]$
3	2048	0.199	$0.201\ [0.194, 0.210]$
4	2048	0.220	$0.211\ [0.191, 0.239]$
3	1024	0.284	$0.263\ [0.247, 0.283]$
3	4096	0.150	$0.160\ [0.155, 0.165]$

pool positive outcomes at $+G$ with negative ones at $-G$ (Appendix B). For $p = 3$ this gives 0.201; the field-grid fit gives 0.199.

The close three-pattern match may come partly from cancelling errors. One pattern removes competition, yet c exceeds c_{CW}; separate one-pattern runs give $c = 1.063\, c_{\mathrm{CW}}$ at $N = 2048$ (Appendix B). The models differ in update order (permutation versus expected sweeps) and interaction strength ($1/N$ versus $1/(N-1)$). Multiplying the $p = 3$ prediction by 1.063 gives 0.212, outside the field-grid interval.

Measured c also falls more slowly with size than predicted: as $N^{-0.36}$ versus $N^{-0.46}$ over $N = 1024$–4096 (two-point slopes, no uncertainties; Eq. (4) gives $N^{-1/2}$ for a fixed ramp). These fixed-schedule comparisons do not measure the rate dependence of Hopfield sampling. Measured and predicted c cross near $N = 2048$, where the schedule was set. Scaled by the one-pattern ratio at each size (1.126, 1.020), the prediction becomes 0.319 at $N = 1024$ and 0.153 at 4096, so the trend remains: measured c is 18% below and 4% above (unrounded values).

Pattern correlation is one untested explanation of the size trend. Random patterns have correlations of order $N^{-1/2}$, which shift the inverse temperatures at which overlaps begin to grow by the same order as the *critical window*, the range over which fluctuations spread the transition at finite N. The shifts need not become negligible relative to this window as N grows (Appendix C).

6. Discussion

The selection law links errors in mode probabilities to the field and noise as the modes form; Ising-chain cooling studies [11] instead track misaligned regions and excess energy.

Annealed importance sampling weights annealed runs to estimate equilibrium averages [9]. An equal-cost comparison with reheating would need valid weights for the filtered start and an assessment of estimator bias and variance. Diffusion models change their distribution gradually, but with an exact score they reverse a noising process and need not equilibrate at each step [4]; extending the calculation needs their dynamics and discrete updates. A variance-preserving noise schedule can miss mode asymmetry; suitable ones recover mode probabilities and within-mode variation in Gaussian mixtures and CW [2].

CW is an exactly computable test case for comparing cooling schedules by their error in mode probabilities; applying c elsewhere requires an equilibrium reference. Measuring Hopfield overlaps during cooling could reveal whether pattern correlations explain the remaining discrepancies.

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Appendix A. Selection formula and exact-chain calculations

From spin updates to drift and noise. For a chosen spin, let $\ell_i = \beta(N-1)^{-1}\sum_{j\neq i} x_j + h$, where $h = \beta G$. Holding the other spins fixed, Eq. (1) assigns weights e^{ℓ_i} and $e^{-\ell_i}$ to its two values, so the redrawn spin has mean $\tanh \ell_i$. For large N, the neighbors' average is approximately m. A uniformly chosen spin has old mean m, so one update changes m by $[\tanh(\beta m + h) - m]/N$ in expectation. With N updates per expected sweep, the drift is $\dot{m} = -m + \tanh(\beta m + h)$. Near $m = 0$ and in a weak field, an update flips a spin with probability about one half, and each flip changes m by $2/N$. The noise variance per expected sweep is therefore approximately $N \times \frac{1}{2} \times (2/N)^2 = 2/N$. A short interval contains many updates that change m little; we approximate the centered sum of their jumps by a Gaussian. Expanding $\tanh z = z - z^3/3 + \cdots$ near $\beta = 1$, retaining the field, linear, and cubic terms, and writing G for $h = \beta G$ (equal at $\beta = 1$) gives

$$dm = \big[(\beta - 1)m - m^3/3 + G\big]\,dt + \sqrt{2/N}\,dW_t, \tag{7}$$

where W_t is a standard Brownian motion. At $G = 0$, Eq. (7) is the swept pitchfork bifurcation analyzed by Berglund and Gentz [3]: one stable value becomes two.

Removing growth without changing the sign. For the linear ramp, $\beta - 1 = \gamma t$. While m is small, omit the cubic term. Without field or noise, $\dot{m} = \gamma t m$ has the positive growth factor $e^{\gamma t^2/2}$. Set $Y_t = e^{-\gamma t^2/2} m_t$ and start at $t_0 < 0$. The product rule gives

$$Y_t = e^{-\gamma t_0^2/2} m_{t_0} + G\int_{t_0}^{t} e^{-\gamma u^2/2}\,du + \sqrt{2/N}\int_{t_0}^{t} e^{-\gamma u^2/2}\,dW_u.$$

The three terms are the remaining initial imbalance, the accumulated field, and the accumulated noise. With $w(u) = e^{-\gamma u^2/2}$, the noise integral is the limit of sums $\sum_j w(u_j)\sqrt{\Delta u}\,\epsilon_j$ of independent standard Gaussians ϵ_j: a Gaussian with mean zero and variance $\sum_j w(u_j)^2 \Delta u$. The field term therefore uses w, and the noise variance uses w^2.

The weights are appreciable over times of order $\gamma^{-1/2}$ around the transition. An early start suppresses the initial term; a late end lets both integrals settle. Extending their limits to the whole real line gives the amplitude Y with

$$\mu_Y = \mathbb{E}Y = G\int_{-\infty}^{\infty} e^{-\gamma u^2/2}\,du = G\sqrt{2\pi/\gamma},$$
$$V_Y = \mathrm{Var}(Y) = \frac{2}{N}\int_{-\infty}^{\infty} e^{-\gamma u^2}\,du = \frac{2}{N}\sqrt{\pi/\gamma}.$$

Hence $U = Y/\sqrt{V_Y} = a + Z$, where $Z \sim \mathcal{N}(0,1)$ and $a = \mu_Y/\sqrt{V_Y} = G\sqrt{N}(\pi/\gamma)^{1/4}$. The positive mode is selected when $U > 0$, with probability $\Pr(Z > -a) = \Phi(a)$, as in Eq. (3). These infinite limits approximate a finite cooling interval; the integrals must settle before the cubic term substantially changes the amplitude.

From a sign probability to retained log-odds. For small a, $\Phi(a) = 1/2 + a/\sqrt{2\pi} + O(a^3)$. The derivative of $\mathrm{logit}\, q$ at $q = 1/2$ is 4, so $\mathrm{logit}\,\Phi(a) = 4a/\sqrt{2\pi} + O(a^3)$. At the final inverse temperature β_f, a field adds $\beta_f N G m$ to each configuration's log weight. To first order in G, the positive mode's log mass changes by its average, $\beta_f N G \bar{m}_f$, and the negative mode's by the opposite

amount. Here $\bar{m}_f$ is the zero-field mean magnetization in the positive mode at β_f. The equilibrium log-odds is thus approximately $2\beta_f NG\bar{m}_f$. Dividing the two leading log-odds gives

$$c \simeq \frac{\sqrt{2}\,\pi^{-1/4}}{\beta_f \bar{m}_f}(N^2\gamma)^{-1/4}.$$

Regime. Over the selection time $T \sim \gamma^{-1/2}$, the linear coefficient γT is of order $\sqrt{\gamma}$, so the cubic term matters once $|m| \sim \gamma^{1/4}$. Noise accumulated over T has typical magnitude $\sqrt{T/N} \sim N^{-1/2}\gamma^{-1/4}$, a fraction $(N\gamma)^{-1/2}$ of this scale. The field contribution must also be small on this scale, $|G|T/\gamma^{1/4} \sim |G|\gamma^{-3/4} \ll 1$, which follows from $N\gamma \gg 1$ if a remains bounded. At equilibrium, Stirling's formula for the $\binom{N}{k}$ configurations with $m = 2k/N - 1$, plus the interaction $N\beta m^2/2$, gives to leading order in N $\log \pi(m) \simeq \text{constant} + N[(\beta-1)m^2/2 - m^4/12]$. At $\beta = 1$ the quadratic terms cancel, so fluctuations reach $|m| \sim N^{-1/4}$; the quadratic term matters when $|\beta - 1| \sim N^{-1/2}$, the critical window of Section 5. Selection spans an inverse-temperature range $\gamma T \sim \sqrt{\gamma}$; requiring it to be wider than this window again gives $N\gamma \gg 1$. These conditions allow linear selection before nonlinear growth fixes the magnitude; they do not exclude later switching.

Exact reduction and numerical evolution. Waiting times between CW updates are independent exponentials with mean $1/N$, so the count k of positive spins is a Markov chain whose rates, at a given temperature, depend only on k. Each spin updates once per expected sweep on average, so each rate is the number of eligible spins ($N - k$ negative or k positive) times their redraw probability:

$$W_k^+ = \frac{N-k}{2}\left[1 + \tanh\left(\beta\frac{2k-N+1}{N-1} + h\right)\right], \tag{8}$$

$$W_k^- = \frac{k}{2}\left[1 - \tanh\left(\beta\frac{2k-N-1}{N-1} + h\right)\right], \tag{9}$$

where $h = \beta G$; the $+1$ and -1 exclude the updated spin from the average of its neighbors. Equal probability flow between adjacent counts, $\pi_k W_k^+ = \pi_{k+1} W_{k+1}^-$ (*detailed balance*), holds for

$$\pi_k \propto \binom{N}{k} \exp\left[\frac{\beta}{2(N-1)}\big((2k-N)^2 - N\big) + h(2k-N)\right], \tag{10}$$

where the binomial factor counts configurations with the same k. The conditional reference is $q_{\text{eq}} = \sum_{k>N/2}\pi_k / \sum_{k\neq N/2}\pi_k$; the omitted zero-magnetization mass is below 10^{-36} at $\beta_f = 1.6$ for $N \geq 512$. We evolve the count probabilities $P_k(t)$,

$$\dot{P}_k = W_{k-1}^+ P_{k-1} + W_{k+1}^- P_{k+1} - (W_k^+ + W_k^-)P_k,$$

by implicit Euler steps with Richardson extrapolation. The result therefore has no sampling error.

Linear ramps. The ramps of Section 3 start in the equilibrium distribution at their field (time steps: Table 5). Table 2 compares the resulting c_{num} with the full-formula prediction $c_{\text{sel}} = \text{logit}\,\Phi(a)/\,\text{logit}\,q_{\text{eq}}$, with a from Eq. (3). The excess over c_{sel} halves when N doubles at $\gamma = 0.004$ and 0.001, but not at 0.00025. We did not derive the correction that causes this excess.

Table 2: Linear-ramp comparison with the full selection formula. Entries give numerical c and, in parentheses, $100(c_{\mathrm{num}}/c_{\mathrm{sel}} - 1)$.

N	$\gamma = 0.004$	$\gamma = 0.001$	$\gamma = 0.00025$
1024	0.09370 (0.93%)	0.13393 (1.81%)	0.18868 (1.03%)
2048	0.06587 (0.46%)	0.09372 (0.97%)	0.13314 (1.23%)
4096	0.04645 (0.23%)	0.06589 (0.49%)	0.09361 (0.86%)

Table 3: **Cooling schedule.** The dwell is $\lceil 6\tau_{\mathrm{des}} \rceil$ sweeps (we chose the multiplier six), where τ_{des} is the zero-field CW inverse spectral gap of the full chain for $\beta \leq 1$ and of the reflected chain above. Counts computed at $N = 2048$ are reused at every size; Hopfield runs apply them as permutation sweeps, the CW evolution as expected sweeps. For the CW evolution at $G = 0.0006125$, halving the time step from 0.02 to 0.01 changes c_{CW} by 2.8×10^{-11}; over the eight fields, c_{CW} ranges from 0.119498 to 0.119808. Multiplying only the final cooling's counts by 0.25 and 4 gives $c = 0.098978$ and 0.146625 (ratio 1.4814).

β	.60	.80	.90	.95	1.00	1.05	1.10	1.20	1.40	1.60
sweeps	15	30	59	107	355	145	65	31	16	11

Fixed schedule. Mirroring the Hopfield protocol, the CW evolution starts with $\mathrm{Binomial}(N, 1/2)$ counts, runs two sweeps at each temperature of Table 3, removes $k \geq N/2$ (including $m = 0$) and renormalizes, reheats for 30 sweeps at $\beta = 0.7$, then applies Table 3. The removal keeps $k < N/2$ for both field signs, as the Hopfield runs remove $(1, +)$ for both, so the two logits in c_{CW} share the sign of G. Switching after selection is unlikely while each dwell is much shorter than the mean time for the exact zero-field chain to pass between mode centers: at $N = 2048$, about $10^{7.7}$, $10^{22.9}$, and $10^{137.8}$ expected sweeps at $\beta = 1.1$, 1.2, and 1.6, against 834 sweeps in the whole schedule. Eq. (4) describes a constant-rate ramp from a start well below $\beta = 1$, whereas this schedule dwells at ten fixed temperatures after a filtered start and reheating; we did not isolate which difference makes sixteenfold final cooling raise c less than twofold (Table 3).

Schedule design. The relaxation time τ is the inverse *spectral gap*, the smallest positive decay rate of the chain at fixed temperature: in time τ, the slowest deviation from equilibrium shrinks by e^{-1}. The six-τ dwell of Table 3 is a relaxation prescription, not a bound on the error in final mode probabilities. To measure relaxation within a sign, the *reflected* chain (even N) keeps $k = N/2, \ldots, N$, sets $W^{-}_{N/2} = 0$, and leaves the other rates unchanged. Below the transition, the schedule uses the full chain, because reflection would remove the slow relaxation of the balance between signs. At $\beta = 1$ and $G = 0$, the full-chain inverse gap is 42.2, 59.0, and 82.7 sweeps for $N = 1024$, 2048, and 4096; it grows as $\sqrt{N}$, so dwells redesigned for each N would too.

Appendix B. Simulation and statistical methods

Sampling and labels. Each update redraws x_i with probability $\Pr(x_i = +1 \mid x_{-i}) = [1 + \tanh(\beta H_i)]/2$, where x_{-i} denotes the other spins and $H_i = N^{-1} \sum_\mu \xi^\mu_i \sum_j \xi^\mu_j x_j - (p/N) x_i + G\xi^1_i$;

Table 4: Complete field grid, pooled over twenty pattern sets per row. Of 40,000 initial runs per row, removing $(1,+)$ leaves 32,467–34,178 to reheat (33,331 at $G=0$); n finish assigned to pattern 1 (ties go to the first pattern index: 101 initial, no final ties), and n_+ of these have positive sign. At $G=0$, the Wilson interval is $[0.4863, 0.5048]$. The last column is the 95% interval of $q_{\text{chain}}(G) + q_{\text{chain}}(-G)$, with the two Wilson half-widths added in quadrature on each side; it does not resample shared pattern sets or adjust for four comparisons. After the reheat, the CW full-chain inverse gap ($N=2048$, $\beta=0.7$, $G=0$) gives $e^{-30/\tau_{\text{full}}} \approx 1.2\times10^{-4}$; this does not bound conditional Hopfield memory.

$10^4 G$	n	n_+	q_{chain}	reference q_{eq}	paired sum
6.125	11085	7417	.6691	.9726	[0.98620, 1.01058]
4.375	11131	7038	.6323	.9275	[0.99624, 1.02136]
2.625	11110	6355	.5720	.8220	[0.98614, 1.01209]
0.875	11033	5783	.5242	.6248	[0.98221, 1.00852]
0	11220	5560	.4955	.5	
−0.875	11080	5221	.4712	.3752	
−2.625	11229	4796	.4271	.1780	
−4.375	11617	4374	.3765	.0725	
−6.125	11801	3886	.3293	.0274	

the term $-(p/N)x_i$ removes self-interaction. All runs begin with independent unbiased spins, and the field is present during the initial cooling, reheating, and final cooling. Only final samples labeled pattern 1 with nonzero sign (*retained* samples) enter the estimate. Every nonzero-field point's sample interval lies strictly between $1/2$ and its reference interval; all eight nonzero fields of Table 4 enter the fit, and none was excluded. Table 5 lists seeds and run counts.

Equilibrium reference. For each pattern set s, 256 runs start at $+\xi^1$ and run at $\beta_f = 1.6$, $G=0$. After 60 equilibration sweeps, overlaps are recorded for 200 sweeps; all measurements were labeled $(1,+)$. Their mean $\bar{m}_s^1$ gives the weak-field reference $\operatorname{logit} q_{\text{eq},s}(G) \approx 2\beta_f N G \bar{m}_s^1$. The standard error (SE) divides the standard deviation of the pattern set's 256×200 overlaps by $\sqrt{256\times200/\tau_{\text{ref}}}$, with $\tau_{\text{ref}} = 1.6946$ the reflected CW relaxation time at $N=2048$, $\beta=1.6$, $G=0$. τ_{ref} is an approximate correlation correction, fixed before the runs, not a measured Hopfield correlation time. The standard error of the 256 independent run means, which needs no correlation time, is 0.85–1.05 times the reported SE for each pattern set. The log-odds envelope adds the statistical term $1.96\,\text{SE}$ and a 1% systematic allowance for the reference, a design choice rather than a measured bound:

$$\Delta L = 2\beta_f N|G|\big[1.96\,\text{SE}(\bar{m}^1) + 0.01\bar{m}^1\big].$$

The pooled references in Table 4 and Figure 2b weight pattern set s by $\omega_s = n_s(G)/\sum_r n_r(G)$, with $n_s(G)$ its retained samples at field G: $\bar{m}^1 = \sum_s \omega_s \bar{m}_s^1$ and $\text{SE}(\bar{m}^1)^2 = \sum_s \omega_s^2 \text{SE}(\bar{m}_s^1)^2$. In Figure 2b, this envelope (both terms) gives the horizontal error bars, and pooled 95% Wilson intervals give the vertical ones. A common 1% increase in the overlaps reduces c by about 1%. The bootstrap excludes the allowance and error in the Hopfield equilibrium approximation. For the CW reference at $N=2048$, $\beta=1.6$, the exact and weak-field q_{eq} differ by at most 2.108×10^{-8} over the eight fields, negligible relative to the observed distortion.

Table 5: Runs and seeds. Pattern set s is drawn with NumPy `default_rng`$(20260823 + 1009s)$. Each Hopfield run uses its own segment of a seeded PyTorch GPU generator. Bootstraps draw 20,000 pattern-set resamples and keep both field signs of a set together.

Experiment	Runs, steps, and seeds
Fig. 1a paths	Noise $\sqrt{2/N}$; 14 paths from $m = 0$ at $N = 1024$; 6 sweeps at each of 360 equally spaced β from 0.7 to 1.35; Euler step 0.02; NumPy seed 4
CW linear ramps	β piecewise constant at the midpoint of each interval $\Delta\beta = 0.005$, held for $0.005/\gamma$ expected sweeps; each interval has $0.25/\gamma$ (rounded) Richardson-extrapolated implicit Euler steps of about 0.02
Field grid	$s = 0$–19; 2000 initial runs per set and field; nine fields including $G = 0$; dynamics and bootstrap seed 20260825
Equilibrium reference	$s = 0$–19; 256 runs per set; dynamics seed 20260823
Pattern-count and size scans	$s = 3$–14; 8000 initial runs per set and field sign; dynamics seed $9000 + p$ (pattern count) or $4000 + N$ (size, $N \neq 2048$); bootstrap seed 2026 (pattern count) or 7 (size)
One-pattern runs	$s = 0$; 150,000 initial runs per field sign; $N = 512$–4096; dynamics seed $31337 + N$

Slope estimate. For field j and pattern set s, let $x_{js} = \operatorname{logit} q_{\mathrm{eq},js}$, $y_{js} = \operatorname{logit} \widehat{q}_{js}$ with $\widehat{q}_{js} = n_{+,js}/n_{js}$, and $[L_{js}, U_{js}]$ the 95% Wilson interval of $\widehat{q}_{js}$ [12]. Each pattern set's slope weights its measurements by the inverse squared logit half-width of this interval:

$$w_{js} = \left[\frac{\operatorname{logit} U_{js} - \operatorname{logit} L_{js}}{2 \times 1.96}\right]^{-2}, \qquad c_s = \frac{\sum_j w_{js} x_{js} y_{js}}{\sum_j w_{js} x_{js}^2}.$$

The grid estimate is $\widehat{c}_{\mathrm{grid}} = \sum_s n_s c_s / \sum_s n_s$, with n_s the retained samples over the set's eight fields. Resampling the twenty pattern sets 20,000 times, with reference means $\bar{m}_s^1$ held fixed, gives $\widehat{c}_{\mathrm{grid}} = 0.1994$ and the 2.5th–97.5th percentile interval $[0.1908, 0.2095]$. Three alternative estimators give 0.1994–0.1997: weights $n_{js}\tilde{q}_{js}(1 - \tilde{q}_{js})$, with $\tilde{q}_{js}$ the Wilson-interval center; all fields pooled before fitting; and a pooled binomial logistic fit $\operatorname{logit} q = \alpha + cx$ (intercept 0.000944).

Pattern-count and size comparisons. With $n_{+,s}(G)$ and $n_s(G)$ the positive and total retained counts of pattern set s at field G, let

$$S_s = n_{+,s}(+G) + n_s(-G) - n_{+,s}(-G), \qquad R_s = n_s(+G) + n_s(-G).$$

The pooled estimator $\widehat{c}_{\mathrm{scan}} = \operatorname{logit}(\sum_s S_s / \sum_s R_s) / \operatorname{logit} q_{\mathrm{eq}}^{\mathrm{CW}}(N)$ uses the CW reference in place of the Hopfield one and does not fit a slope per pattern set. The $p = 3$, $N = 2048$ row of Table 1 comes from separate runs on twelve of the field grid's pattern sets ($s = 3$–14). One-pattern runs ($s = 0$; full protocol with the filtered start; $|G| = 0.0006125$) give c equal to 1.065, 1.126, 1.063, and 1.020 times $c_{\mathrm{CW}} = 0.227$, 0.171, 0.120, 0.090 at $N = 512$, 1024, 2048, 4096.

Appendix C. Amplification and pattern correlations

Conditioning on the winning pattern. Under the independent-Gaussian approximation, let $m^1 \sim \mathcal{N}(\delta, \sigma^2)$, where δ is the field-induced mean shift and the other overlaps have zero mean and variance σ^2. The density f of m^1 gives $f(r)/f(-r) = \exp(2r\delta/\sigma^2)$ at magnitude $r > 0$. Winning depends only on the magnitude, so it does not change these sign odds at fixed r:

$$\Pr(m^1 > 0 \mid |m^1| = r, \text{ pattern 1 wins}) = \frac{1}{1 + e^{-2r\delta/\sigma^2}} \simeq \frac{1}{2} + \frac{r\delta}{2\sigma^2}.$$

At $\delta = 0$, all patterns have the same distribution, so the winning magnitude is distributed as σM_p, where $M_p = \max_{1 \le \mu \le p} |Z_\mu|$ for independent standard Gaussians Z_μ. The magnitude distribution is even in δ, so its first-order change is zero. Averaging the sign probability over winning magnitudes then gives, to first order,

$$q_{\text{chain}} \simeq \frac{1}{2} + \frac{\delta}{2\sigma}\,\mathbb{E}M_p.$$

Expanding the logit around $1/2$ gives

$$\operatorname{logit} q_{\text{chain}} \simeq \frac{2\delta}{\sigma}\,\mathbb{E}M_p.$$

With one pattern, $M_1 = |Z|$. Dividing by this one-pattern response, with the same equilibrium-log-odds reference, gives Eq. (6). The calculation averages sign probabilities before taking log-odds. It gives sign odds given the winning pattern, not all pattern-pair frequencies.

Two patterns. For $p = 2$, the largest miss in Table 1, setting $y_i = \xi_i^1 x_i$ splits the model exactly into two non-interacting CW blocks of 980–1068 spins, whose signs agree exactly when pattern 1 wins. Treating the blocks as independent and equal doubles the logit 0.343384 of the one-pattern runs at $N = 1024$; dividing by the exact CW $\operatorname{logit} q_{\text{eq}} = 3.575$ at $N = 2048$ gives $c = 0.19212$, against the measured 0.18783 $[0.184, 0.191]$ and the Gaussian prediction 0.169. The block sizes and initialization remain unmatched.

Pattern correlations. For p patterns, $C_{\mu\nu} = N^{-1} \sum_i \xi_i^\mu \xi_i^\nu$ records their pairwise correlations. For continuous-time random-site dynamics at zero field, averaging the expected Gibbs redraw over sites, weighted by ξ_i^μ, gives

$$\dot{m}^\mu \simeq -m^\mu + \frac{1}{N} \sum_i \xi_i^\mu \tanh\left(\beta \sum_\nu \xi_i^\nu m^\nu\right) \simeq -m^\mu + \beta \sum_\nu C_{\mu\nu} m^\nu.$$

The first approximation neglects the self-exclusion correction of order p/N; the second uses $\tanh z \simeq z$ for small overlaps. The drift of $\boldsymbol{m} = (m^1, \ldots, m^p)$ is thus $(\beta C - I)\boldsymbol{m}$. Along an eigenvector of C with eigenvalue λ, an overlap grows at rate $\beta\lambda - 1$, beginning at $\beta = 1/\lambda$ rather than at $\beta = 1$ as for $C = I$. Each off-diagonal entry averages N independent random signs, so its standard deviation is $N^{-1/2}$. For fixed p, the eigenvalue shifts and the resulting transition shifts have this order, comparable to the critical window of Section 5. Separating these shifts from the selection range $\gamma T \sim \sqrt{\gamma}$ would require $N\gamma \gg 1$. This comparison does not identify the cause of the measured size trend.